This code runs a simulation in which you are able to interact with an interface. The image describes the following: a slice of the spinal cord of a macaque, an cylindrical electrode and 500 simulated neurons. In this simulation, the macaque is completely paralysed, meaning that no natural activity will occur in the neurons. In order to activate them, an electromagnetic field must be present. The further neurons lat from the electrode, the weaker the electromagnetic field is and the less likely they are to be activated. In the orange area, there are points that are known to correlate with specific muscle movements (hence the muscle ID). The code “punishes” the program when it unnecessarily uses current and/or activates neurons you do not wish to activate. This program is able to determine the optimum for the x, y and current this way. The user is able to submit the muscle they wish to stimulatie in the macaque, and the program will determine the optimum. Note that there are limitations to this program, for example not all the blue parts of the picture (where neurons are) are close enough to these datapoints to determine to what muscle group they belong.